

Roll No.

--	--	--	--	--	--	--

END SEM EXAMINATION-2025

Name of the Program: B.Tech. (Biotech.)

Semester: II

Name of the Course: Engineering Mathematics

Paper Code: TMA-202

Time: 03 Hours

Maximum Marks: 100

Note:

- (i) Answer **all the questions** by choosing *any two of the sub questions*.
- (ii) Each question carries 10 marks.

Q1		(20 marks)	
(a)	Define rank of a matrix and find the rank of the matrix $A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ 2 & 13 & 10 \end{bmatrix}$.		CO1/CO2
(b)	Solve the following system of linear equations: $x + y + z = 6$; $x + 2y + 3z = 14$; $-x + y - z = -2$.		
(c)	Find the Eigen values and Eigen vectors of the matrix: $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$		
Q2		(20 marks)	
(a)	Define the continuity of a function and check the continuity of the given function: $f(x) = \begin{cases} 2, & x \leq 0 \\ 3x + 2, & 0 < x \leq 1 \\ \frac{x}{x-1}, & x \geq 1. \end{cases}$		CO2/ CO3
(b)	Find the maximum or minimum value of the following function: $y = x^3 - 9x^2 + 24x - 12$.		
(c)	Find gradient, divergence and curl of $\vec{v} = (xyz)\hat{i} + 3x^2y\hat{j} + (xz^2 - y^2z)\hat{k}$ at point (2, -1, 1).		
Q3		(20 marks)	
(a)	Evaluate the following limits:		

	(i) $\lim_{x \rightarrow 0} \frac{x}{\sqrt{2+x} - \sqrt{2-x}}$ (ii) $\lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{x^2}$.	CO3/CO4
(b)	Define mean value theorem and verify it for the function $f(x) = 4x^3 - 8x^2 + 7x - 2$ on $[2, 5]$.	
(c)	Define Taylor's and Maclaurin's series and evaluate the first four terms of the Taylor's series for $f(x) = x^3 - 10x^2 + 6$ at $x=3$.	
Q4	(20 marks)	CO4/ CO5
	Evaluate the following integrals:	
(a)	(i) $\int (3x^2 + 2x + 6)dx$, if $y=5$ when $x=0$ and (ii) $\int \log x \, dx$.	
	Find dy/dx of the following:	
(b)	(i) $y = \frac{4-x^2}{4+x^2}$ and (ii) $y = \frac{1+e^x}{1-e^x}$.	CO5/ CO6
(c)	Define Beta and Gamma functions and evaluate the integral $\int_0^{\infty} e^{-x} x^5 \, dx$.	
Q5	(20 marks)	
	Find the area bounded by the x axis and the curve $y = x^2$ between $x = 1$ and $x = 2$.	
(a)		
(b)	Define sequence and series with examples and examine the nature of the given series: $1+2+3+\dots+n+\dots\infty$.	
(c)	Define Fourier series and find the Fourier series for $f(x) = x, 0 < x < 2\pi$.	